A GAN-Based Sparse Recovery Approach for Airborne Radar Clutter Suppression

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***Abstract*—The strong extended clutter caused by high-speed moving platforms is a key factor that degrades the detection performance of slow-moving targets in radar systems. Space-Time Adaptive Processing (STAP) techniques can estimate the clutter covariance matrix and thereby achieve adaptive clutter suppression, leading to effective target detection. However, the performance degradation of STAP due to the lack of independent and identically distributed (i.i.d.) training samples remains a critical issue to be addressed. In this paper, a sparse-regularized Generative Adversarial Network (GAN) is proposed to generate additional synthetic samples by learning the spatial-temporal information from existing clutter echo data. Through the adversarial training process between the generator and discriminator, the neural network learns the amplitude distribution characteristics of clutter space-time cells and reconstructs the clutter covariance matrix accordingly. Finally, simulation experiments demonstrate the feasibility of the proposed method.**

Keywords—Clutter suppression, sparse recovery, generative adversarial network

I. INTRODUCTION

Airborne radar systems inevitably encounter strong clutter signals in addition to target returns and thermal noise during operation. The high-speed movement of the platform induces Doppler spread in clutter echoes from stationary ground scatterers. As a result, this type of clutter may exhibit Doppler characteristics similar to those of weak slow-moving targets, leading to a significant degradation in target detection performance. To address this issue, Space-Time Adaptive Processing (STAP) [1] has been widely adopted to suppress clutter by estimating its 2-order statistical properties and generating adaptive filters in both spatial and temporal domains, thereby substantially improving the radar's ability to detect low-velocity targets.

This work was supported in part by the National Natural Science Foundation Program of China under Grants 62501401, supported by Shanghai QiYuan Innovation Foundation, in part by the USCAST2023-29 and SAST2024-026. (Corresponding Author: Penghui Huang)

At the same time, STAP also faces several challenges. The performance of STAP is closely related to the estimation accuracy of the clutter covariance matrix. When the clutter exhibits significant range dependence, the estimation of the covariance matrix suffers from notable errors, which can severely degrade the performance of clutter suppression.

To address this issue, various methods for estimating the clutter covariance matrix have been proposed in recent years. These approaches aim to improve the accuracy of covariance estimation by reducing the number of clutter samples required [2], [3] or compensating for nonstationary errors [4], [5]. In this paper, for the first time, a Generative Adversarial Network (GAN) is explored to assist in the estimation of the clutter covariance matrix.

The GAN [6] was introduced by Ian Goodfellow et al. in 2014 and has since become a major direction in machine learning. It learns data distributions through a competition between two networks: the generator and the discriminator. During training, the discriminator evaluates real and generated samples, updating its weights to improve classification. The generator then adjusts its parameters to better fool the discriminator. This iterative process continues until equilibrium is reached, at which point the generator produces realistic samples and the discriminator can no longer reliably distinguish them from real data. Due to its powerful generative ability, GAN has been widely applied in image generation, data augmentation, and style transfer.

Aiming at the problem of limited training samples under the independent and identically distributed (i.i.d.) clutter assumption, this paper proposes a sparse-regularized GAN. During training, the generator and discriminator compete adversarially to learn the spatial-temporal amplitude characteristics of clutter in the range cell under test. This enables accurate reconstruction of the clutter covariance matrix and thereby enhances clutter suppression performance.

Finally, simulation experiments are conducted to validate the effectiveness of the proposed network.

II. METHODOLOGY

A. Clutter Signal Model

The geometric configuration of the clutter model for an airborne monostatic radar is illustrated in Fig. 1, where all relevant parameters are labelled. For a given clutter cell i, the corresponding elevation and azimuth angles are denoted as $\theta_{el,i}$ and $\theta_{azi,i}$, respectively. d_{azi} represents the antenna element spacing along the azimuth direction, and M and N denote the number of elements in the elevation and azimuth directions, respectively. V_a is the velocity of the airborne platform, and θ_{yaw} is the angle between the platform's velocity vector and the azimuth direction of the antenna array.

Correspondingly, the spatial frequency and the normalized Doppler frequency are, respectively, given by [7]

$$f_{azi,i} = \frac{-d_{azi} \sin\theta_{el.i} \cos\theta_{azi,i}}{\lambda}, \quad f_{t,i} = \frac{2V_a \sin\theta_{el,i} \cos\left(\theta_{azi,i} - \theta_{yaw}\right)}{\lambda PRF} \tag{1}$$

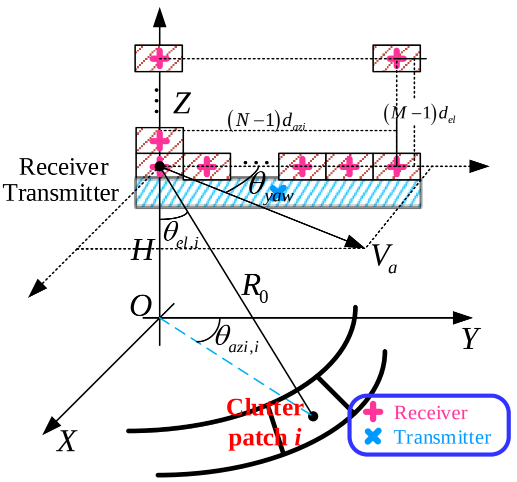

Fig. 1 3-D geometric model of an airborne radar system.

Then according to (1), the corresponding space-time steering vector can be written by

$$\begin{aligned} \boldsymbol{s}_i &= \boldsymbol{s}_{t,i} \otimes \boldsymbol{s}_{azi,i}, \\ \boldsymbol{s}_{azi,i} &= \left[1, e^{2\pi f_{azi} j}, e^{2\pi\times 2 f_{azi} j} \cdots, e^{2\pi(N-1) f_{azi} j}\right]^T, \\ \boldsymbol{s}_{t,i} &= \left[1, e^{2\pi f_{t,i} j}, e^{2\pi\times 2 f_{t,i} j} \cdots, e^{2\pi(K-1) f_t j}\right]^T \end{aligned} \tag{2}$$

The clutter signal received in range gate R_i can be written by

$$\boldsymbol{x}_c = \sum_{i=1}^{N_c} A_i \boldsymbol{s}_i \tag{3}$$

where N_c represents the number of clutter cells and A_i represents the clutter amplitude coefficient, which can be calculated by the radar equation as well as clutter fluctuation distribution.

Due to the high-speed motion of airborne radar systems, a higher pulse repetition frequency (PRF) is often employed in practice to reduce Doppler ambiguity. However, this inevitably leads to more severe range ambiguity. Therefore, the clutter power increases accordingly as follows:

$$\boldsymbol{x}_c = \sum_{r=1}^{N_r} \sum_{i=1}^{N_c} A_{i,r} \boldsymbol{s}_{i,r} \tag{4}$$

where N_r represents the number of range ambiguity.

2.2 Proposed GAN-Based Sparse Recovery Approach

This subsection provides an overview of the constructed deep convolutional GAN (DCGAN). The discriminator is implemented as a multi-layer convolutional neural network [8], which is used to determine whether the input data is generated by the generator. Taking the case where both the number of channels and pulses are 8 as an example, Fig. 2 presents the schematic structure of the discriminator. The input data consists of two channels, with the first and second channels representing the real and imaginary parts of the original complex-valued 8×8 data, respectively. After passing through four convolutional layers, the feature maps are flattened into a fully connected layer of 512 dimensions, and finally passed through a sigmoid activation function to produce a probability value in the range of [0, 1].

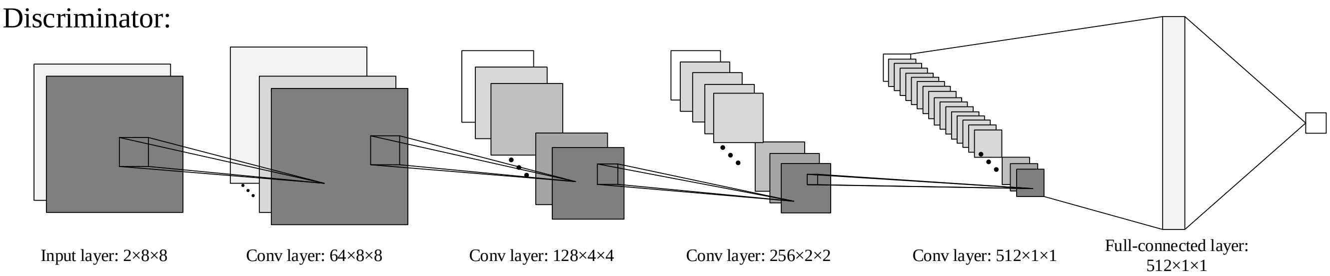

Fig. 2. The schematic structure of the discriminator.

In the adversarial network, the configuration of the generator is shown in Fig. 3, which relies on a precomputed sparse dictionary. The steering vectors corresponding to the spatial frequency f_s and temporal (Doppler) frequency f_t are given by

$$\begin{aligned} \boldsymbol{s}_t(f_t) &= \left[1 \quad \exp(j2\pi f_t) \quad \cdots \quad \exp\left[j2\pi f_t (K-1)\right]\right]^T, \\ \boldsymbol{s}_s(f_s) &= \left[1 \quad \exp(j2\pi f_s) \quad \cdots \quad \exp\left[j2\pi f_s (N-1)\right]\right]^T \end{aligned} \tag{5}$$

Considering that the convolutional layers in the discriminator can only extract local features from two-dimensional data, and time-domain signals are not capable of locally representing frequency-domain information, it is necessary to adopt two-dimensional data in the frequency domain as atoms of the sparse dictionary. The atom corresponding to the space-time frequency pair $\left(f_s, f_t\right)$ can be expressed as:

$$\boldsymbol{s}_F\left(f_s, f_t\right)=\mathcal{F}_2\left[\boldsymbol{s}_s\left(f_s\right)\times\boldsymbol{s}_t^T\left(f_t\right)\right] \tag{6}$$

where $\mathcal{F}_2(\ )$ represents the 2-dimensional Fourier transform operation.

Then the whole sparse atom dictionary can be written by

$$\boldsymbol{\Phi}=\left\{\boldsymbol{s}_F\left(f_s, f_t\right)\middle| \begin{array}{l} f_s=-0.5,-0.5+\dfrac{1}{N_s},\cdots,-0.5+\dfrac{N_s-1}{N_s}; \\ f_t=-0.5,-0.5+\dfrac{1}{N_t},\cdots,-0.5+\dfrac{N_t-1}{N_t} \end{array}\right\} \tag{7}$$

The amplitude set can be defined as

$$\boldsymbol{\Sigma}=\left\{\sigma\left(f_s, f_t\right)\middle| \begin{array}{l} f_s=-0.5,-0.5+\dfrac{1}{N_s},\cdots,-0.5+\dfrac{N_s-1}{N_s}; \\ f_t=-0.5,-0.5+\dfrac{1}{N_t},\cdots,-0.5+\dfrac{N_t-1}{N_t} \end{array}\right\} \tag{8}$$

where σ_i represents the signal amplitude of the *i*-th steering vector in $\boldsymbol{\Phi}$, which are learnable parameters of the GAN. N_t and N_s are the numbers of partitions along the spatial and Doppler frequency dimensions of the dictionary. The generating formulation of the generator can be denoted by

$$\text{Output}=\sum_{f_s, f_t} z\left(f_s, f_t\right)\sigma\left(f_s, f_t\right)\boldsymbol{s}_F\left(f_s, f_t\right) \tag{9}$$

where $z\left(f_s, f_t\right)$ is the input random noise, which corresponds to the fluctuation of the clutter amplitude.

So far, we have constructed both the generator and the discriminator of the GAN, In the following, we design the corresponding loss functions for network training. The total loss of the discriminator can be expressed as

$$L_D=-\frac{1}{2}E_{x\sim p_{data}(x)}\left[\log D(\boldsymbol{x})\right]-\frac{1}{2}E_{z\sim p_z(z)}\left\{\log\left[1-D\left(G(z)\right)\right]\right\} \tag{10}$$

where $p_{data}\left(\boldsymbol{x}\right)$ represents the distribution function of the true clutter data, $p_z\left(z\right)$ represents the distribution function of the input noise data and $D(x)$ represents the score assigned by the discriminator to the input data, representing the estimated probability that the input is a real sample. The score lies within the range [0, 1].

The goal of the generator is to deceive the discriminator into believing that the generated data is real. Accordingly, the loss function of the generator is designed to maximize the misclassification rate of the discriminator on the generated samples. That is, to encourage the discriminator to assign a high probability to fake samples. The loss function of the generator can be defined as

$$L_G=-\frac{1}{2}E_{z\sim p_z(z)}\left[\log D\left(G(z)\right)\right] \tag{11}$$

Additionally, considering the sparsity of clutter data, a sparsity regularization term is incorporated into the loss function to enforce sparsity in the learned representations.

$$L_{G-\text{Regularization}}=\lambda\left\|\boldsymbol{\Sigma}\right\|_1 \tag{12}$$

Finally, the loss function of the generator with a sparsity regularization term can be written by

$$L_{G-\text{SR}}=L_G+L_{G-\text{Regularization}} \tag{13}$$

So far, we have completed the construction of the Generative Adversarial Network with sparse regularization. As illustrated in Fig. 4, the training process of the GAN is demonstrated schematically. By training the network on the dataset, the parameters of the generator can be learned effectively, enabling accurate reconstruction of the clutter spectrum and subsequent clutter suppression.

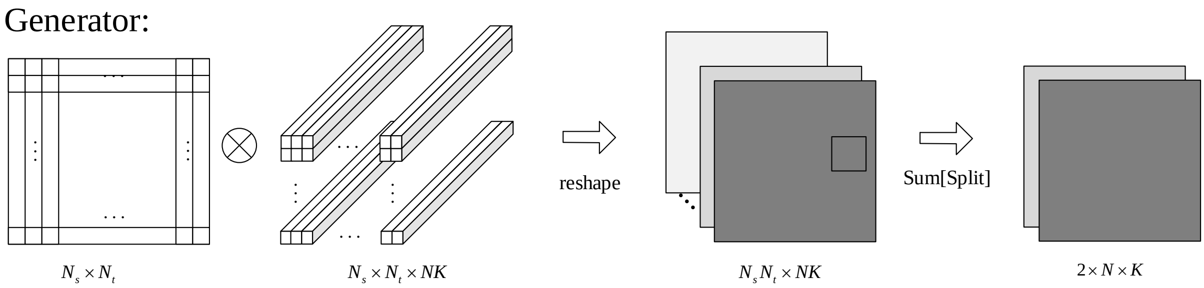

Fig. 3. The schematic structure of the generator.

III. Results

The radar system parameters used in this experiment are summarized in Table 1. The simulation dataset consists of 60,000 clutter samples generated based on these radar parameters. The loss curves obtained during the training of GAN, as described in the previous section and shown in Fig. 5, indicate that after multiple training epochs, both the generator and discriminator losses gradually converge to stable values. Unlike other neural networks where the loss typically decreases monotonically, in GANs both the generator and discriminator update their parameters through adversarial learning, and the loss does not necessarily decrease throughout the training process.

Fig. 6 presents a 2D image of the learned sigma parameters after network training. It can be observed that, similar to the space-time spectrum of a side-looking airborne radar, the values exhibit high peaks along the diagonal, while remaining close to zero in other regions due to the imposed sparsity constraint.

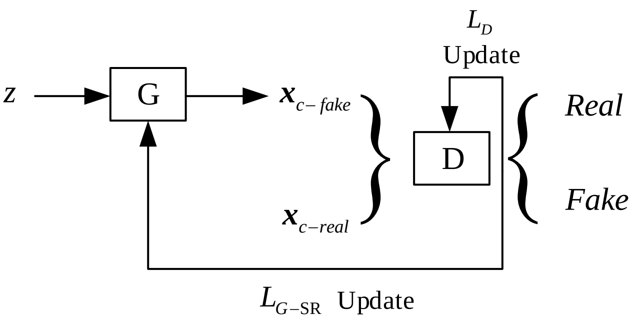

Fig. 4. The schematic structure of GAN.

TABLE 1. PARAMETERS OF AN AIRBORNE RADAR SYSTEM

wavelength	0.3m
Pulse repetition frequency	2000Hz
Pulse number	8
Channel number	8
Platform velocity	150m/s
Clutter to noise ratio	60dB

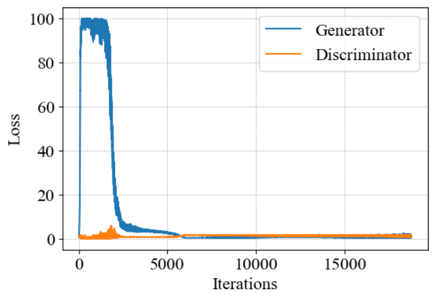

Fig. 5. The loss curves of the generator and discriminator.

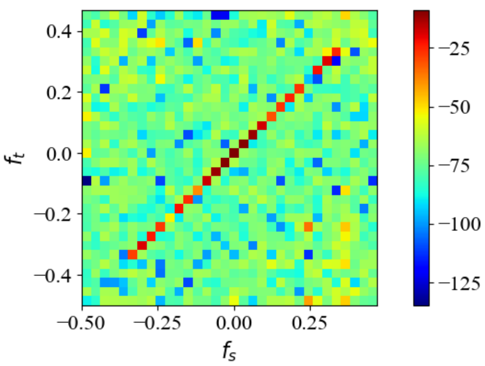

Fig. 6. 2D image of the learned sigma parameters after network training.

After reconstructing the clutter covariance matrix, the finally estimated space-time spectrum is obtained, as shown in Fig. 7. Except for some errors near the edges, the estimated spectrum closely approximates the ideal clutter space-time distribution, which can be further validated by the SCNR (Signal-to-Clutter-plus-Noise Ratio) loss curves depicted in Fig. 8.

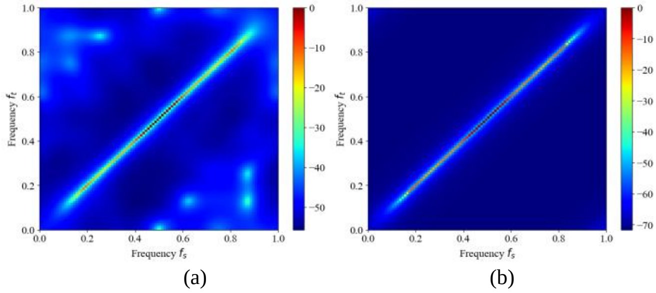

Fig. 7. (a) Reconstructed space-time spectrum through proposed method. (b) ideal space-time spectrum.

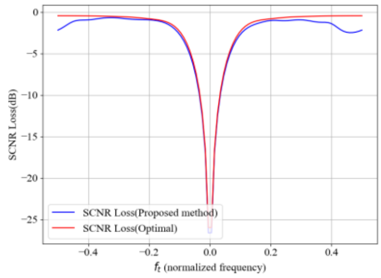

Fig. 8. Comparison of the SCNR loss curves between the proposed method and the optimal method.

IV. CONCLUSION

In this study, a sparse-regularized Generative Adversarial Network (GAN) is developed for space-time spectral estimation, covariance reconstruction, and clutter suppression. The simulation results validate the capability of the proposed GAN in reconstructing the space-time clutter spectrum with high accuracy. This work presents a promising and novel application of deep learning techniques in the field of radar clutter suppression, offering new insights into data-driven approaches for adaptive signal processing in a high-speed motion platform.

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